INCH-POUND

MIL-PRF-1/796C
23 July 1999
SUPERSEDING
MIL-E-1/796B
8 July 1977

PERFORMANCE SPECIFICATION SHEET

# ELECTRON TUBE, GAS SWITCHING
# TYPE 6024

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: ATR, waveguide type, F = 2,800 MHz, peak incident power 1.0 kw.

ABSOLUTE RATINGS:

| Parameter:<br>Unit: | Transmitter Po<br>kw | TA<br>°C | Alt<br>ft |
|---|---|---|---|
| Maximum:<br>Minimum: | 1,000<br>20 | +100<br>-40 | 10,000<br>--- |

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Mounting position: Any 2/

TEST CONDITIONS:

| Parameter:<br>Unit: | Transmitter Po<br>kw | tp<br>µs | Du<br>--- | VSWR<br>--- |
|---|---|---|---|---|
| Test 1:<br>Test 2: | 20<br>750 | 1.0 ±10%<br>1.0 ±10% | 0.001<br>0.001 | 1.05<br>1.20 |
| Test frequency: | F = 2,800 ± 5.0% | | | |

See footnotes at end of table I.

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GENERAL:

Qualification: Required.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Testing and inspection.

| Inspection | Method | Test | Conditions | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | | | | Min | Max | |
| Qualification inspection | | | | | | | |
| Degradation due to vibration | 4021 | --- | | --- | --- | --- | --- |
| Loaded Q | 4461 | --- | 2/ | QL | --- | 5.5 | --- |
| Conformance inspection, part 1 | | | 4/ | | | | |
| Tuning susceptance | 4482 | ---- | 2/ | b | -0.05 | +0.05 | --- |
| Firing time | 4486 | 1 | 2/ 3/ | t | --- | 10 | sec |
| Arc loss | 4488 | 1 | 2/ 3/ | La | --- | 0.8 | dB |
| Temperature cycling (nonoperating) | 1027 | --- | 1/ | --- | --- | --- | --- |
| Temperature cycling life test | 1027 | --- | Group C 1/ | --- | 50 | --- | Hz |
| Conformance inspection, part 2 | | | | | | | |
| High-level VSWR | 4474 | 1 | | --- | --- | 1.15 | --- |
| Normalized conductance | 4484 | --- | 2/ | g | --- | 0.05 | --- |
| Conformance inspection, part 3 | | | | | | | |
| Life test | 4551 | 2 | Group D 2/ | t | 1,000 | --- | hrs |
| Life-test end points: | --- | --- | | | | | |
| Tuning susceptance | 4482 | --- | 2/ | b | -0.05 | +0.05 | --- |
| Normalized conductance | 4484 | --- | 2/ | g | --- | 0.1 | --- |
| Firing time | 4486 | 1 | 2/ 3/ | t | --- | 10 | sec |

1/ Satisfactory test on any one of 1B44, 1B52, 1B53, 1B56, 1B57, 5921, or 5922 ATR tube types made during the same two week period, shall be evidence that all of the group meet this test requirement.

2/ Mount in accordance with Drawing 153-JAN.

3/ The tube shall be mounted as in 2/ and followed by a matched load. The tube shall fire within the limits specified after application of rf power. This test shall be performed at least 7 days after pumping and at least 24 hours after any previous discharge.

4/ Unless otherwise specified, the acceptance level for all tests listed under conformance inspection, part 1, shall be 1.0 percent, inspection level II.



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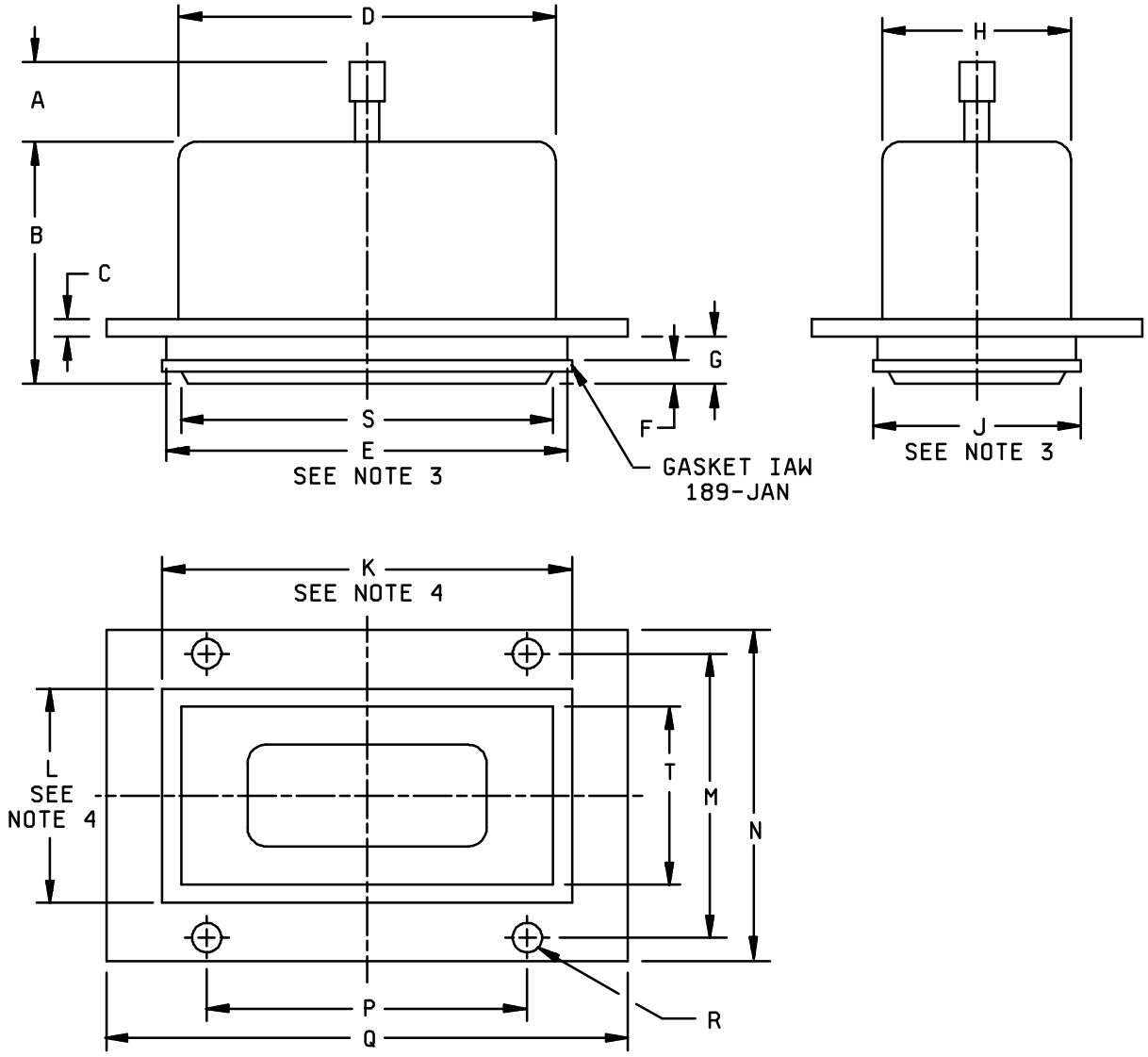


NOTES:

1. Dimensions are in inches.
2. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Dimensions "E" and "J" pertain to flange.
4. Dimensions "K" and "L" pertain to outside of gasket.
5. Dimension "F" shall be measured prior to attachment of gasket.
6. Gasket to be securely mounted to tube.

FIGURE 1. Outline drawing of electron tube type 6024.



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| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| Conformance inspection, part 1 | | | | |
| A | --- | .625 | --- | 15.88 |
| B | --- | 1.875 | --- | 47.62 |
| C | .109 | .141 | 2.77 | 3.58 |
| D | 2.985 | 3.015 | 75.82 | 76.58 |
| F | .165 (see note 5) | .195 (see note 5) | 4.19 | 4.95 |
| G | .347 | .377 | 8.81 | 9.58 |
| H | 1.486 | 1.516 | 37.74 | 38.51 |
| K | --- | 3.223 | --- | 81.86 |
| L | --- | 1.725 | --- | 43.82 |
| N | 2.609 | 2.641 | 66.27 | 67.08 |
| Q | 4.109 | 4.141 | 104.37 | 105.18 |
| R | .203 DIA (4 holes) | .245 DIA (4 holes) | 5.16 DIA (4 holes) | 6.22 DIA (4 holes) |
| S | 2.990 | 3.010 | 75.95 | 76.45 |
| T | 1.490 | 1.510 | 37.85 | 38.35 |
| Conformance inspection, part 2 | | | | |
| E | 3.203 | 3.233 | 81.36 | 82.12 |
| J | 1.703 | 1.733 | 43.26 | 44.02 |
| M | 2.150 | 2.350 | 50.60 | 59.69 |
| P | 2.400 | 2.600 | 61.06 | 66.04 |

FIGURE 1. Outline drawing of electron tube type 6024 - Continued.



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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR
Navy - AS, CG, MC, OS, SH
Air Force - 99

Preparing activity:
DLA - CC

Project (5960-3551-08)



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